



**Michigan
Technological**
University

Ferris Wheel Final Report

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L14 Team 2

Evaluating Ferris Wheel Baseline Inertia

Table 1. Period, frequency, test inertia, and total inertia of all weight models with their respective three tests

	No Weight Added			200 Grams Added			300 Grams Added			600 Grams Added			Wheel Without Load		
Test #	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3
Period (s)	1.392	1.394	1.390	1.686	1.680	1.682	1.814	1.808	1.809	2.146	2.1147	2.145	4.866	4.880	4.860
Frequency (Hz)	0.718	0.717	0.719	0.593	0.595	0.594	0.5551	0.554	0.553	0.466	0.466	0.466	0.206	0.205	0.206
Test Inertia (kg·m ²)	N/A			0.00313			0.00469			0.00938			0.07677		
Total Inertia (kg·m ²)	0.00686			0.00998			0.0115			0.0162			0.0835		

The Ferris Wheel rotational inertia was evaluated using a dynamic rotating inertia test rig that behaves as a single degree of freedom rotational spring–mass system. Free vibration tests were performed by rotating the table and releasing it, and the natural frequency and period were extracted from the measured angular response. Tests were conducted for the unloaded table, the table with two, three, and six calibration masses placed symmetrically at a known radius, and the unloaded Ferris Wheel, with three trials completed for each configuration. Calibration inertias were calculated analytically using a point-mass approximation, $I = \sum mr^2$, and are reported alongside the measured natural frequency and period values. Added inertia was plotted versus the square of the natural period for all calibration mass cases (12 total data points), and a linear fit was applied. The fit equation and coefficient of determination (R^2) were included on the plot, and the slope and intercept were used to determine the table stiffness and baseline table inertia, respectively. The Ferris Wheel inertia was then calculated by substituting its measured natural periods into the fitted relationship, yielding three inertia values. The average and standard deviation of these values were reported to quantify measurement repeatability, with all results presented using appropriate units.

Table 2. Wheel moment of inertia for all trials along with the mean and standard deviation

	Trial 1	Trial 2	Trial 3	Mean	Standard Deviation
Wheel Moment of Inertia (kg·m ²)	0.0766	0.0771	0.0764	0.0767	0.0035

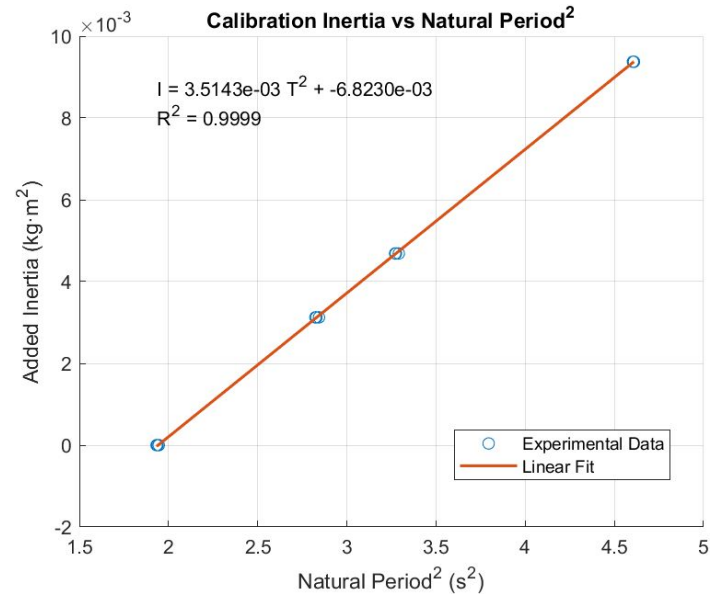


Figure 1. Linear fit of added inertia over natural period of 12 points of corresponding experimental data

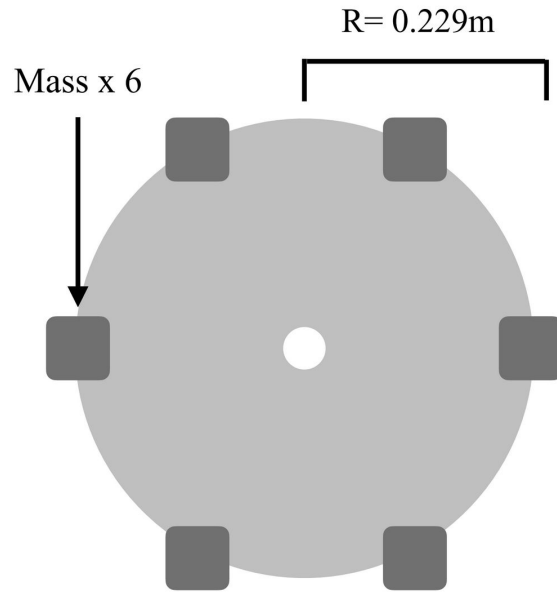


Figure 2. Model of ferris wheel with radius and masses

The mass moment of inertia of the Ferris Wheel was calculated for each loading configuration in Section 4.2 using the parallel axis theorem. The baseline inertia was taken as the average from dynamic testing. For loaded configurations, the total inertia was the sum of the baseline inertia and the contributions from added masses in the carts. Figure 2 shows a model of the ferris wheel. Each added mass was modeled as a point mass at the radial distance from the rotation axis to the cart center. Inertia contributions were calculated using $I=mr^2$ and summed across all six carts. Required dimensions, including radial distance and cart mass, were defined in a schematic. The total inertia for each configuration was then obtained by adding the calculated added inertia to the baseline value.

Table 3. Each ferris wheel configuration with their corresponding added inertia

Configuration	Added Inertia ($\text{kg}\cdot\text{m}^2$)
No Mass	0
100g Per Cart	3.14×10^{-4}
200g Per Cart	6.28×10^{-4}

Calculated inertias for the baseline, 100 g per cart, and 200 g per cart cases were tabulated in Table 3 for use in subsequent analysis of angular response, torque, power, and kinetic energy.

Assumptions and limitations include measurement noise and linear fit errors in the baseline test, treating added masses as point masses, assuming symmetric cart placement, and neglecting friction, damping, and structural flexibility. These factors may cause small discrepancies between calculated and true system inertia, especially for higher loadings.

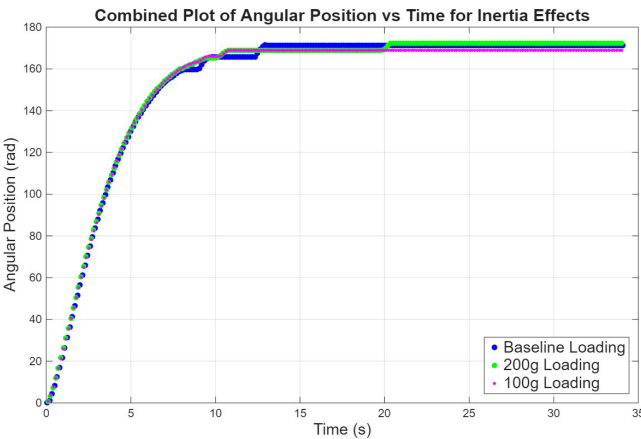


Figure 3. Combined plot of angular position for one test case from each loading configuration for proportional controlled case

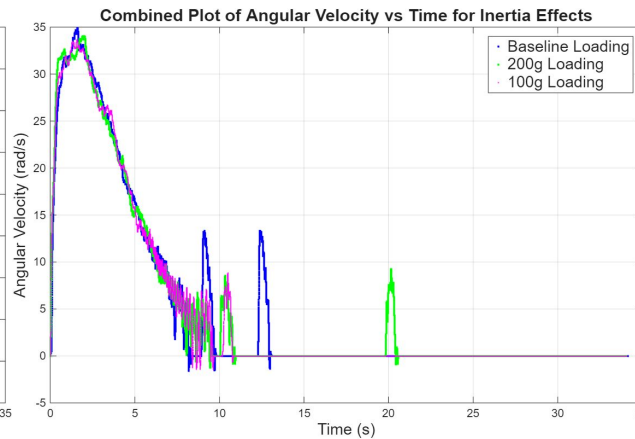


Figure 4. Combined plot of angular velocity for one test case from each loading configuration for proportional controlled case

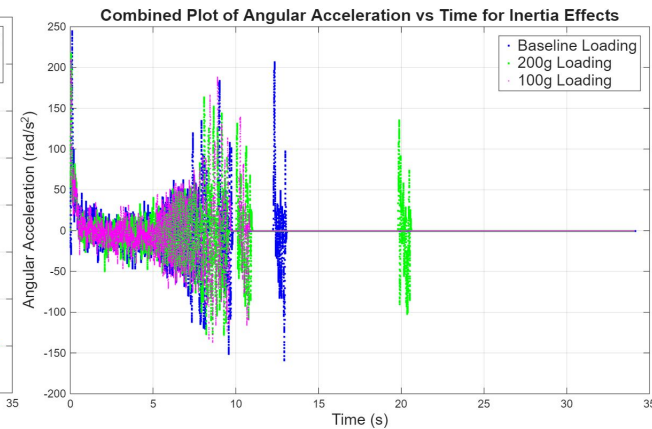


Figure 5. Combined plot of angular acceleration for one test case from each loading configuration for proportional controlled case

Table 4. The peak angular velocity, acceleration, mean, and standard deviation of repeat test cases

Test case	Angular Velocity (rad/s)	Angular Acceleration (rad/s ²)	Mean Angular Velocity (rad/s)	Mean Angular Acceleration (rad/s ²)	Standard Deviation Angular Velocity (rad/s)	Standard Deviation Angular Acceleration (rad/s ²)
Baseline 1	33.54	273.23	33.75	264.24	1.05	16.61
Baseline 2	34.89	245.08				
Baseline 3	32.82	274.41				
100g 1	33.78	189.09	33.68	208.62	0.09	18.70
100g 2	33.65	210.39				
100g 3	33.60	226.37				
200g 1	33.74	162.80	33.88	185.53	0.15	28.33
200g 2	34.03	217.26				
200g 3	33.88	176.52				

The angular position curves for each test case are similar in shape, with the 100g loading being the smoothest curve and the baseline loading being the roughest curve as shown in Figure 3. Angular velocity curves for each test case follow the same curve of a sharp initial increase then a decrease with unsteady changes in velocity as the ferris wheel approaches a stop as shown Figure 4. Angular acceleration gradually increases in variation as time continues as shown in Figure 5. There are spikes in acceleration and velocity which correspond to the P gain correcting the final position of the ferris wheel. Table 4 shows test to test variability of peak angular velocity and acceleration for all tests. All tests have a similar value for peak velocity however, the acceleration varies between tests, with some tests having a much greater acceleration. Following the same trend as the individual tests, the peak velocity for each test configuration is similar but the acceleration varies under different loading configurations.

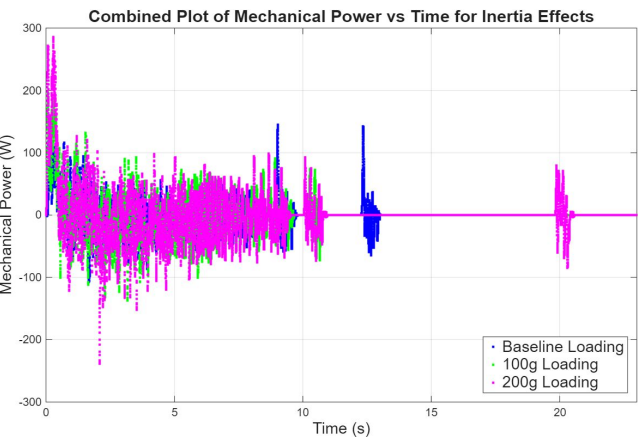


Figure 6. Combined plot of mechanical power for one test case from each loading configuration for proportional controlled case

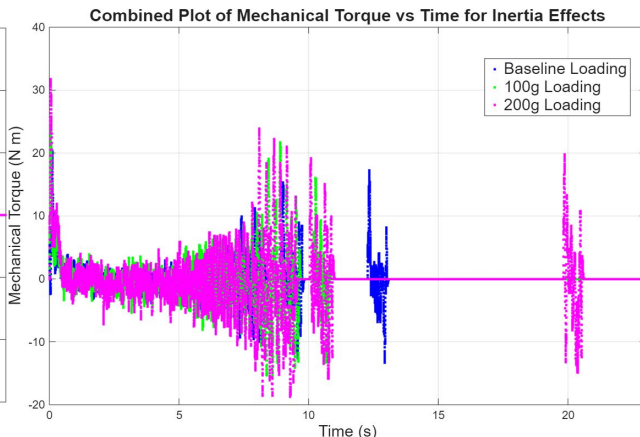


Figure 7. Combined plot of mechanical torque for one test case from each loading configuration for proportional controlled case

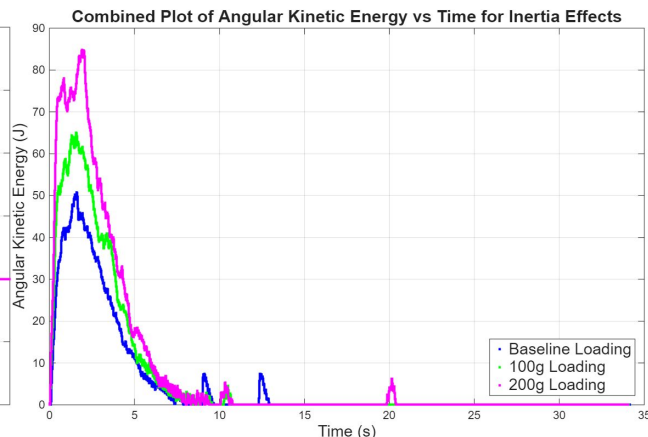


Figure 8. Combined plot of angular kinetic energy for one test case from each loading configuration for proportional controlled case

Table 5. The peak mechanical power, torque, angular kinetic energy (KE) mean, and standard deviation of repeat test cases

Test case	Mechanical Power (W)	Mechanical Torque (N m)	Angular KE (J)	Mean Power (W)	Mean Torque (N m)	Mean Angular KE (J)	Standard Deviation Power (W)	Standard Deviation Torque (N m)	Standard Deviation Angular KE (J)
Baseline 1	233.53	22.81	46.97	202.53	22.06	47.59	39.75	1.39	2.98
Baseline 2	157.71	20.46	50.83						
Baseline 3	216.36	22.91	44.96						
100g 1	193.26	21.75	65.61	207.07	23.99	65.21	20.52	2.15	0.36
100g 2	197.30	24.19	65.10						
100g 3	230.65	26.03	64.91						
200g 1	224.05	23.83	83.32	251.23	27.16	84.04	31.37	4.15	0.073
200g 2	285.56	31.81	84.78						
200g 3	244.09	25.84	84.03						

Mechanical power was calculated by multiplying torque and angular velocity, Figure 6 shows the power calculated for each loading configuration. Power for each loading configuration is similar, with a roughly constant range of power while the ferris wheel is in motion. Mechanical torque was calculated by multiplying loading configuration moment of inertia by angular acceleration. Figure 7 shows the torque calculated for each loading configuration. Torque for all cases start at a small range and gradually increase as the ferris wheel approaches its final position. Angular kinetic energy (KE) was calculated by multiplying one half by loading configuration mass moment of inertia by angular velocity. As seen if figure 8, KE for each configuration follows the same curve with peak KE increasing with mass loaded which is shown with values in Table 5. Table 5 also shows that power and torque increase with mass loaded because the 200g configuration has the greatest values while the baseline configuration has the smallest values. Increasing inertia increases the required mechanical torque and power to reach the same angular position, and KE increases directly with inertia. Discrepancies can come from the P control which changes the velocity of the ferris wheel, changing the torque, power, and KE as a result.

Impact of Inertia on Ferris Wheel Angular Response - PI Control

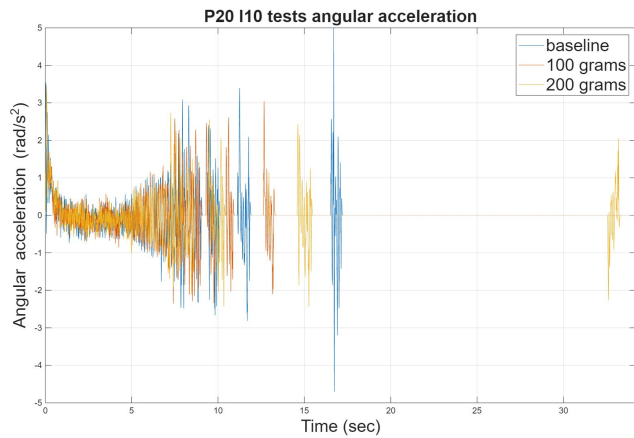


Figure 9: combined plot of angular acceleration for each test case from optimized gain P20I10

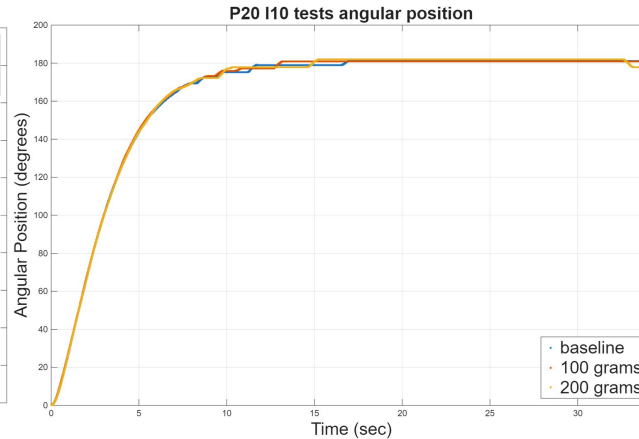


Figure 10: combined plot of angular position for each test case from optimized gain P20I10

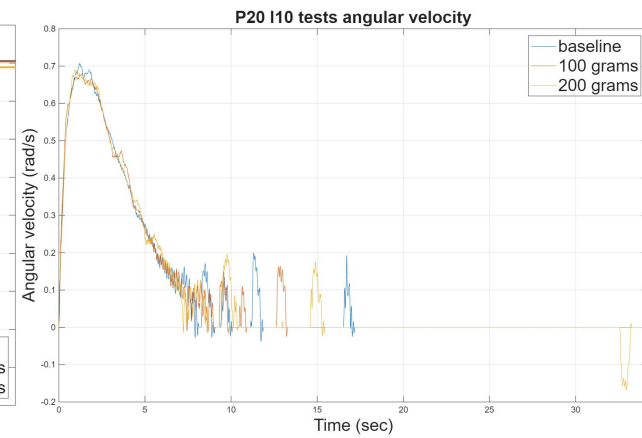


Figure 11: combined plot of angular position for each test case from optimized gain P20I10

When observing data for the chosen PI gain while under different loads there are some noticeable trends. The peak angular velocity decreased by about 0.01 rad/s per 100 grams added, while the peak angular acceleration would decrease by about 0.07 rad/s² with the initial 100 grams, and had virtually no decrease with 200 grams.

Using PI gains has benefits over just P gain, mainly it helps decrease steady state error as the ferris wheel system tries to approach 180 reference angle. This is supported when comparing plots of the two systems, where the P20 system falls short of the 180 degree angle, while the P20I10 system reaches the 180 degrees. Since the P gain is the same between the two tests increased inertia is handles relatively the same as the system doesn't have any additional power to apply.

Table 6: P20I10 peak angular velocity and acceleration for all test cases

Test case	Peak Angular Velocity (rad/s)	Peak Angular Acceleration (rad/s ²)
Baseline	0.7069	3.52846
100 grams	0.6918	3.4509
200 grams	0.6811	3.4497

Impact of Inertia on Ferris Wheel Mechanical Torque, Power and Angular Kinetic Energy - PI Control 6

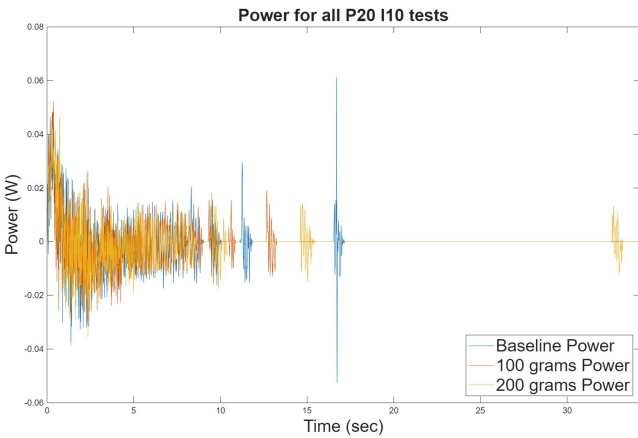


Figure 12: combined plot of Power for each test case from optimized gain P20I10

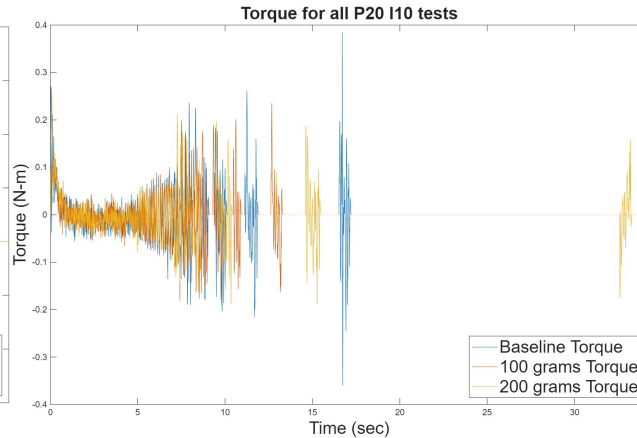


Figure 13: combined plot of Torque for each test case from optimized gain P20I10

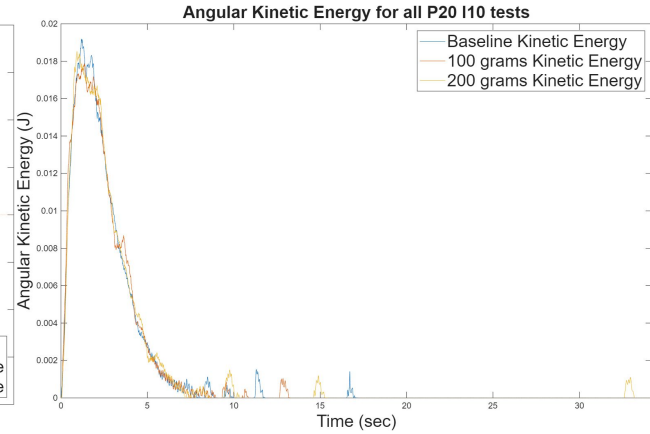


Figure 14: combined plot of angular kinetic energy for each test case from optimized gain P20I10

When the inertia of the ferris wheel system is increased it has an effect on torque, power, and kinetic energy. As seen in table 7 with the first 100g there is an initial drop in peak torque of about 0.12 N-m, peak power of about 0.011W, and peak kinetic energy of about 0.002 J. When 200 grams were introduced changes in peak torque was negligible, while peak power dropped by about 0.01W and peak kinetic energy actually increased by 0.001 J

When comparing plots of P20 and P20I10 the power and torque plots appear to have little to no change. However the P20I10 angular kinetic energy seem to be much more consistent across the different weight tests compared to the P20 tests. This is most likely due to the I gain calling for a higher or lower rotational speed as needed allowing its Kinetic Energy to be more controlled.

Table 7: P20I10 peak torque, power, and kinetic energy for all test cases

Test case	Mechanical Torque (N-m)	Power (W)	angular kinetic energy (J)
Baseline	0.3836	0.0613	0.0192
100 grams	0.2660	0.0521	0.0179
200 grams	0.2670	0.0464	0.0185

Critical Design Review (CDR) of Ferris Wheel Engineering Project

7

Pic from MEPH Instructors, Canvas

Table 8. Customer Requirements

Customer Requirement:	Specific Target:
Time to move 180 degrees	Settling time ≤ 16 seconds
Positional Accuracy	Steady State Error of ≤ 1 degree of target
Control System	Use Standard PID Controller
Software	Use Beaglebone Blue and Matlab
Rider comfort	Angular Acceleration $\leq 4 \text{ rad/s}^2$
Rider Comfort	Angular Velocity $\leq 8 \text{ rad/s}$

The requirements shown in Table 8 were given by the customer and converted to specific engineering specifications to be measured against.

Apart from the two requirements specifying to use the hardware already planned, these requirements were met by repeated testing of the existing ferris wheel model with varying proportional and integral (PI) gain. The recorded data was processed using Matlab to ensure the requirements were met after testing.

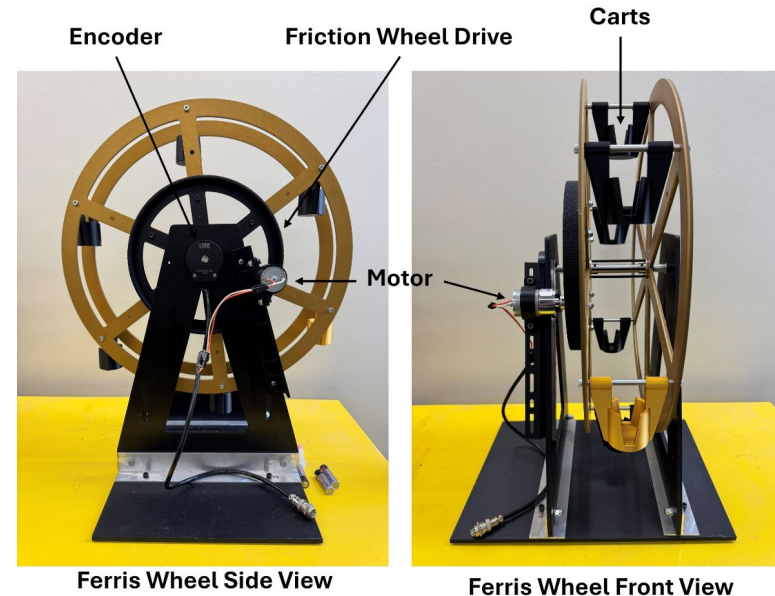


Figure 15. Test Ferris Wheel with Components Labeled

Figure 15 shows the Ferris Wheel model used to test the varying P and I gains. Not shown is the microcontroller used to control the system and communicate with Matlab to run the trials.

This model was controlled using an existing Matlab graphical interface program used to modify PI gain values and export the position, current, voltage values with respect to time. After each trial, the Matlab data was exported to a separate script for processing.

A similar Matlab script was used to simulate the ferris wheel, exporting similar data but without the cost of a physical model.

Critical Design Review (CDR) of Ferris Wheel Engineering Project Cont. 8

Table 9. Significant Values From the P20I10 Trial

Torque (N-m)	0.0403
Power (W)	0.0069
Angular Kinetic energy (J)	0.0046

Table 9 shows the average torque, power, and angular kinetic energy for our chosen PI gain. These values are based on testing our model and would need to be scaled up with the mass and efficiency of the real case.

Table 10 shows a list of significant values relevant to the requirements derived from the test for our chosen PI gain. Each of these values are within the requirements, with the closest being the peak angular acceleration at 0.239 off the maximum. This is necessary to create an enjoyable ride and minimize settling time.

Table 10. P20I10 Trial Requirement Values

Final Angle (deg)	179.828
Overshoot (deg)	-0.172
Settling Time (s)	12.590
Peak Angular Velocity (rad/s)	0.670
Peak Angular Acceleration (rad/s ²)	3.761

Pic from MEPII Instructors, Canvas

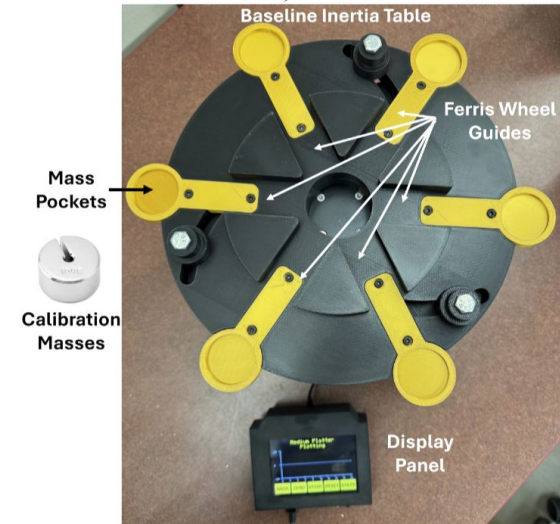


Figure 16. Inertia Test Rig

The inertia testing rig shown in Figure 16 was used to measure the moment of inertia in the loaded versus unloaded ferris wheel frame in order to determine the mechanical power, torque, and angular kinetic energy during use. This was done by first calibrating the inertia rig using known masses of 100g, then testing with an unloaded ferris wheel frame. This allowed us to calculate the torque, power, and angular kinetic energy as well as to measure a baseline moment of inertia for the frame to be added to analytically with the weight of passengers.